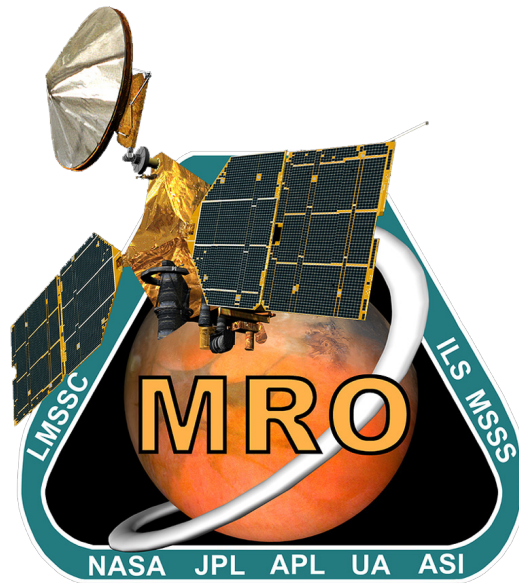




MRO Reverse Engineering Study

Assignment #7

Team 13

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Contents

1	On Board Data Handling Subsystem	1
1.1	Architecture and Rationale	1
1.1.1	Software	1
1.1.2	Hardware	2
1.2	Data Budget	3

1 On Board Data Handling Subsystem

The On Board Data Handling (OBDH) is one of the most important subsystems of the Mars Reconnaissance Orbiter (MRO), as the mission is characterized by its unprecedented data acquisition rates as well as the relay function of the spacecraft (S/C) as part of the Deep Space Network (DSN). The OBDH subsystem is the core of the S/C's avionics, with the responsibility of connecting the incoming datastream of the S/C instruments with the Tracking, Telemetry and Telecommand Subsystem (TTMTC) to transmit data to ground stations. When the downlink is not available, the OBDH subsystem needs to be able to locally store the data until the next scheduled downlink pass. The subsystem thus receives telecommands, schedules their processing, gathers and packages telemetry, and maintains time synchronization across all subsystems.

1.1 Architecture and Rationale

The architectural design of MRO's OBDH subsystem adopts a centralized philosophy, with the central On Board Computer (OBC) governing data handling tasks, and a secondary cold-redundant unit standing by to ensure continuous availability in the event of faults. This architectural simplification helps reducing system complexity and eases integration efforts across the spacecraft's hardware and software layers. Redundancy is applied not only at the processor level but extends to data buses and critical memory interfaces through a cross-strapped configuration. Such a choice ensures that, even under hardware failure conditions, system continuity and responsiveness are preserved [10]. Together, Figure 1 and Figure 2 illustrate both the conceptual and technical layers of MRO's OBDH subsystem. The first shows the cyclic model of science planning and execution, while the second details the encapsulation, storage, transmission, extraction, and archival of scientific data. These images underscore the mission's commitment to data integrity, responsiveness, and traceability throughout its entire lifecycle.

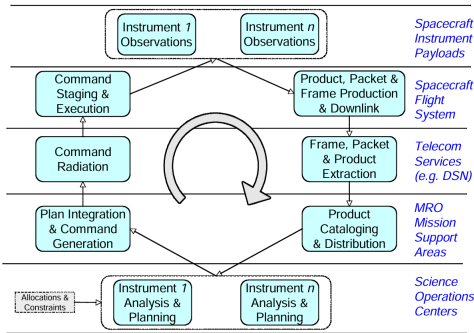


Figure 1: MRO command flow cycle [11]

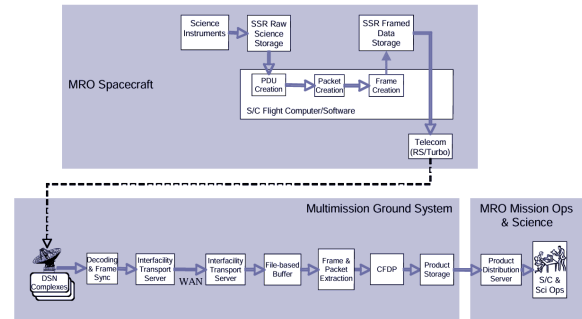


Figure 2: Telemetry data flow [11]

1.1.1 Software

The OBDH software of the MRO plays a central role in managing the spacecraft's high-volume science telemetry. At the software level, the OBDH system adheres to the Consultative Committee for Space Data Systems (CCSDS) standards, structuring its data processing via a layered protocol stack. This begins at the level of Packet Telemetry and Packet Telecommand standards, which define how data is organized and interpreted. For framing and segmentation, the Advanced Orbiter Systems (AOS) transfer frames protocol is employed, supporting the simultaneous handling of multiple data streams while maintaining frame synchronization and minimizing transmission error impacts. To ensure end-to-end delivery of science products, the CCSDS File Delivery Protocol (CFDP) protocol is applied in unacknowledged mode, a choice that significantly reduces overhead and latency, particularly in high-bandwidth applications [6]. Further extending the software's robustness, the OBDH incorporates mechanisms such as the Encoded Event Data Accountability Table (EEDAT), enabling detailed

tracking of data products through all stages, from onboard acquisition to post-downlink processing. The File Delivery Manager (FDM) complements this functionality by managing data packaging, handling partial retransmissions, and maintaining temporal coherence across fragmented datasets. Fault detection and response are tightly interlinked with these tools; health monitoring routines run persistently in the background, monitoring telemetry metrics and responding autonomously when anomalies or inconsistencies are detected. This high degree of autonomy is vital, especially during intervals of delayed ground contact [16]. Indeed, the on-board software does not rely on product-level acknowledgments; instead, it uses a streamlined approach to support unacknowledged CFDP mode, enabling time-critical data handling with minimal protocol overhead. This architecture allows the MRO flight system to autonomously manage, prioritize, and transmit a continuous stream of science data under the constraints of limited bandwidth and variable downlink opportunities [7, 11]

1.1.2 Hardware

Processor The OBC on MRO consists of the X2000 RAD750 processor, a radiation-hardened evolution of the commercial PowerPC 750 architecture, operating at 133 MHz. This processor, capable of executing up to 48 million instructions per second, supports both spacecraft subsystem control and science payload data handling [18]. The use of a single-board architecture simplifies integration, while also enabling robust fault detection and recovery through layered watchdogs and software health monitoring. The RAD750, chosen for its deep space reliability, resists more than 100 krad of radiation and prevents single event latch-ups, crucial for Mars missions. It runs VxWorks flight software, a modular system managing core functions, payloads, and fault protection [18]. This last module constantly scans telemetry, triggers autonomous recovery when possible, or shifts the spacecraft to safe mode pending ground support vital during long Earth-Mars delays [18, 16]. However, its architecture simplifies the integration of both science and housekeeping functions within a unified software environment. Its robust watchdog infrastructure monitors process health continuously, with layered fail-safes capable of initiating soft resets or even triggering a complete swap to the redundant unit in case critical issues arise. This approach to fault mitigation, relying heavily on autonomous software handling, reflects a deliberate design philosophy prioritizing operational continuity.

Memory and Storage MRO makes use of a Solid-state Data Recorder (SSDR) supplied by the company SEAKR Engineering [17], which has a 160 Gbit data storage capability, and degrades to 100 Gbit at end-of-mission. The SSDR consists of an array of 720 Hitachi synchronous DRAM cards with each 256 Mb memory size [20]. The total capacity is divided into two different areas of memory; one dedicated to the raw science data from the instruments, the other specifically for the AOS framed data awaiting downlink. To ensure the instruments can operate independently from each other, the raw science area is further hard partitioned, ensuring that each instrument has enough space for its raw science data. The writing of the data on the SSDR is done by the instrument software itself, while the reading and processing is dedicated to the S/C's Flight Software (FSW). Only OBC and its software are allowed to frame, manage, and queue telemetry products. The packages are eventually transferred to the Small Deep Space Transponder (SDST) for downlink, adhering to a tightly controlled first-in-first-out (FIFO) scheduling logic that guarantees timely and ordered delivery. The other half of the SSDR was first configured as two hard partitions, one for both the X-band and Ka-band downlink. However, since the Ka-band failure, the whole half of the SSDR is set up for the X-band, which includes further smaller soft partitions for each of the instruments. The computer also has available 128 Mb of high-speed, random-access memory [12, 15]

Assuming that the agreement between instrument operators is based on the total data acquisition volume per instrument, the percentage of SSDR would be as described in Table 1. Engineering, Spacecraft Planet Instrument C-matrix Events (SPICE), Mars Climate Sounder (MCS), tracking (for gravity science) and accelerometer data are not counted because of their small volumes.

Instrument	Allocation	Data Volume [Tbits]
CRISM	30%	7.8
CTX	13%	3.5
HiRISE	35%	9.1
MARCI	7%	1.6
SHARAD	15%	4.0

Table 1: MRO data volume allocations.

Bus Architecture Communications within the spacecraft are managed through a combination of robust and high-speed data buses chosen for their specific roles. The command and telemetry backbone employs the MIL-STD-1553B bus standard, chosen for its reliability, deterministic timing, and dual redundancy capability. This bus operates at 1 Mbps, which is sufficient for system control, mode transitions, and low-volume data streams. All main subsystems and instruments interface with this bus as Remote Terminals (RTs), responding to polling sequences initiated by the OBC acting as the Bus Controller [19]. However, due to the limited bandwidth of 1553B, science instruments with high data output, such as HiRISE and SHARAD, utilize dedicated high-speed links to transfer large volumes of data directly to the SSDR. These links are likely to be implemented using RS-422 or LVDS serial protocols, which offer better throughput and electrical robustness. This dual-path bus approach allows control traffic to remain isolated from payload data, preventing congestion, and ensuring reliability across mission phases.

Telecom Interface The final stage in the data chain involves transferring the framed science telemetry from SSDR to the SDST, responsible for modulating and transmitting data through the high-gain antenna of the spacecraft. Here, the MIL-STD-1553B bus is again used for control and configuration tasks, while the high-speed telemetry data path is likely managed via a dedicated high-throughput channel. The SDST supports data rates up to 6 Mbps, making it suitable for MRO's ambitious science return volumes [1]. Redundant cross-strapping is also implemented, allowing either of the two OBC units to interface with the SDST and thus maintaining mission continuity in the event of failure on the OBC side.

1.2 Data Budget

MRO's data budget, divided per subsystem, is shown in Table 2. The following results were estimated by comparison with previous missions or by assumptions based on typical values. In particular, the sampling frequencies were estimated based on the changing rate of the measured parameters; data types were chosen according to typical spacecraft telemetry encoding standards; number of values, n , were selected based on the number of components, and their respective outputs [2, 3, 4, 5, 8, 9, 14]. Regarding the Payloads (P/Ls), the data rate has been estimated based on the total amount of data generated during the Primary Science Phase (PSP) phase, as shown in Table 1, and then averaged over the duration of the PSP. The total estimated data rate is 0.409 Mbps, slightly below the typical operational range floor of 0.5 Mbps [9, 13], that's likely due to omitted parameters or underestimated payload contributions. Assuming a 4-hour storage window between downlinks [9], this yields a total data volume between downlink events of 5.892 Gb, increased to 23.568 Gb with a conservative 400% margin. It is important to note that the mass memory capacity of MRO is 160 Gb [20], which comfortably exceeds the estimated requirement. While this may initially suggest an oversized design, it is not the case: the calculated data volume is based on a highly approximate and conservative lower-bound estimate of the actual data rate and there are periods, such as solar conjunctions, when communications with Earth are completely suspended for weeks and all collected data must be stored locally, justifying the large memory size.

Parameter name	n.	data type	bit size	frequency [Hz]	data rate [bps]
Electric Power Subsystem (EPS)					
Solar array voltage	1	uint16	16	1	16
Solar array current	1	uint16	16	1	16
Battery voltage	1	uint16	16	1	16
Battery current	1	uint16	16	1	16
Bus voltage	1	uint16	16	1	16
Battery temperature	2	sint16	16	0.5	16
On Board Data Handling (OBDH)					
OBC temperature	1	sint16	16	1	16
OBC voltage	1	uint16	16	1	16
OBC current	1	uint16	16	1	16
Watchdog timer	1	uint8	8	0.1	0.8
Boot counter	1	uint8	8	0.1	0.8
Packet count in memory	1	uint16	16	0.01	0.16
Attitude and Orbit Control Subsystem (AOCS)					
Star tracker output	3	float64	64	1	192
Gyro output	3	float32	32	10	960
Sun sensor raw output	24	sint16	16	5	1920
Estimated Sun vector	3	float32	32	2	192
Estimated quaternion	4	float32	32	1	128
ADCS mode	1	uint8	8	0.1	0.8
IMU current	1	sint16	16	1	16
IMU voltage	1	sint16	16	1	16
IMU temperature	1	sint16	16	0.5	8
Propulsion Subsystem (PS)					
Tank pressure	2	uint16	16	0.1	3.2
Tank temperature	2	uint16	16	0.1	3.2
Valve status	4	bool	1	0.1	0.4
Tracking, Telemetry and Telecommand Subsystem (TTMTC)					
Transponder status	2	uint8	8	0.2	3.2
Link error flag	2	bool	1	0.1	0.2
Thermal Control Subsystem (TCS)					
Radiator temperatures	8	uint16	16	0.1	12.8
Heaters status	4	bool	8	0.1	3.2
Payload (P/L)					
HiRISE	-	raw	-	-	141946.2
SHARAD	-	raw	-	-	60834.1
CRISM	-	raw	-	-	121668.2
MARCI	-	raw	-	-	28389.2
CTX	-	raw	-	-	52722.9

Table 2: MRO data budget per subsystem.

-: unavailable.

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Acronyms

AOCS	Attitude and Orbit Control Subsystem. 4
AOS	Advanced Orbiter Systems. 1, 2
CCSDS	Consultative Committee for Space Data Systems. 1
CFDP	CCSDS File Delivery Protocol. 1, 2
CRISM	Compact Reconnaissance Imaging Spectrometer for Mars. 4
CTX	Context Camera. 4
DSN	Deep Space Network. 1
EEDAT	Encoded Event Data Accountability Table. 1
EPS	Electric Power Subsystem. 4
FDM	File Delivery Manager. 2
FIFO	first-in-first-out. 2
FSW	Flight Software. 2
HiRISE	High Resolution Imaging Science Experiment. 3, 4
LVDS	Low-Voltage Differential Signaling. 3
MARCI	Mars Color Imager. 4
MCS	Mars Climate Sounder. 2
MRO	Mars Reconnaissance Orbiter. 1–3
OBC	On Board Computer. 1–3
OBDH	On Board Data Handling. 1, 4
P/L	Payload. 3, 4

PS Propulsion Subsystem. 4

PSP Primary Science Phase. 3

RTs Remote Terminals. 3

S/C spacecraft. 1, 2

SDST Small Deep Space Transponder. 2, 3

SHARAD SHallow Subsurface RADar. 3, 4

SPICE Spacecraft Planet Instrument C-matrix Events. 2

SSDR Solid-state Data Recorder. 2, 3

TCS Thermal Control Subsystem. 4

TTMTC Tracking, Telemetry and Telecommand Subsystem. 1, 4